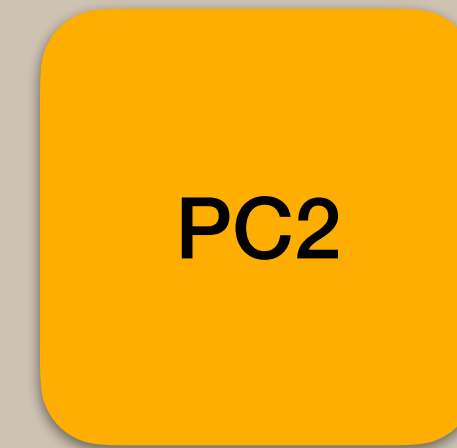
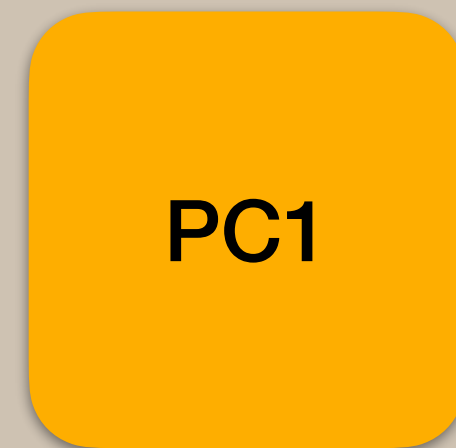


Chapter 2

Local Area Network

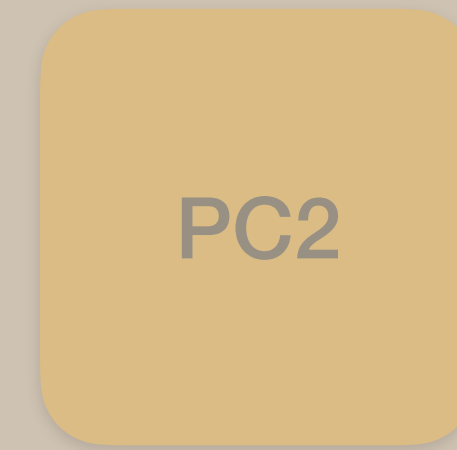
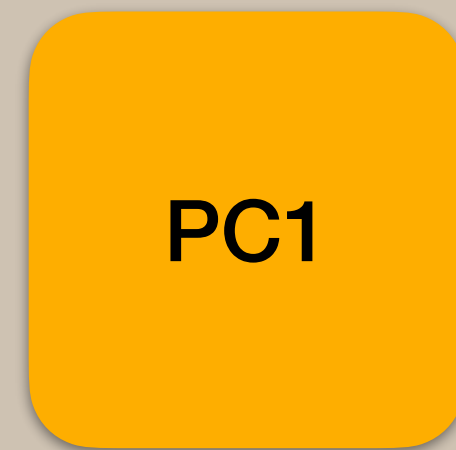
Local Area Network (LAN)

Collection of end devices connected together via network switches



Local Area Network (LAN)

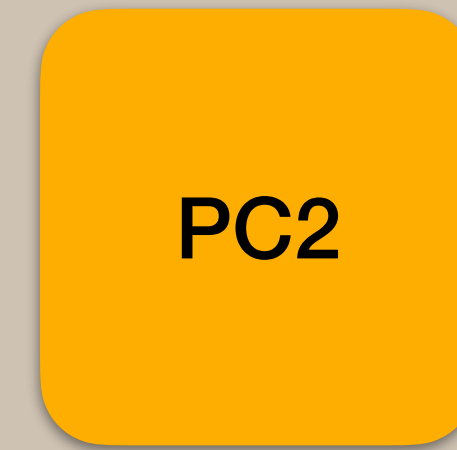
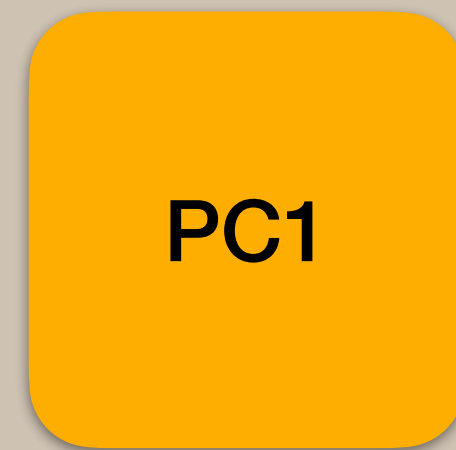
Collection of end devices connected together via network switches



PC1 POV

Local Area Network (LAN)

Collection of end devices connected together via network switches



PC2 P0V

Local Area Network (LAN)

Collection of end devices connected together via network switches



Peer to Peer Network

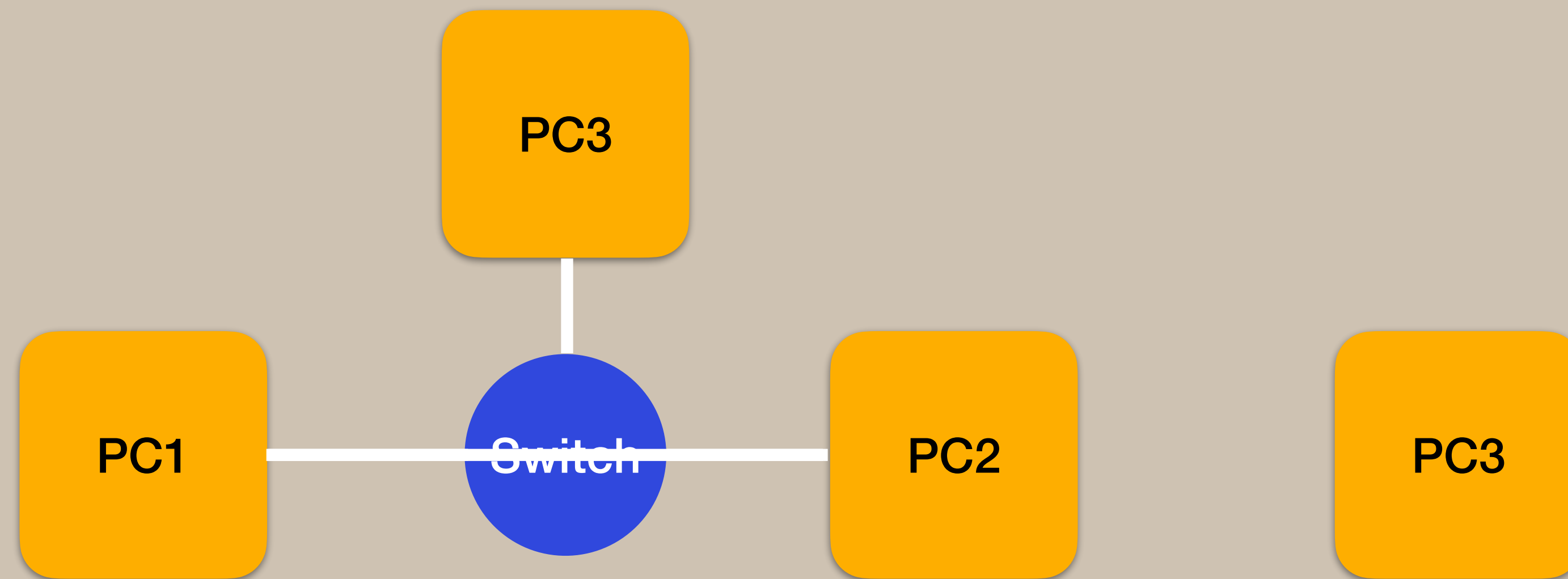
Local Area Network (LAN)

Collection of end devices connected via network switches

A local area network (LAN) is a group of computers and other devices that are connected together in a limited geographical area, such as a home, office, or building. These devices are connected using wired or wireless technology and can share resources such as printers, files, and internet connections.

LANs are typically used to provide network access to a small group of users and are not connected to the internet.

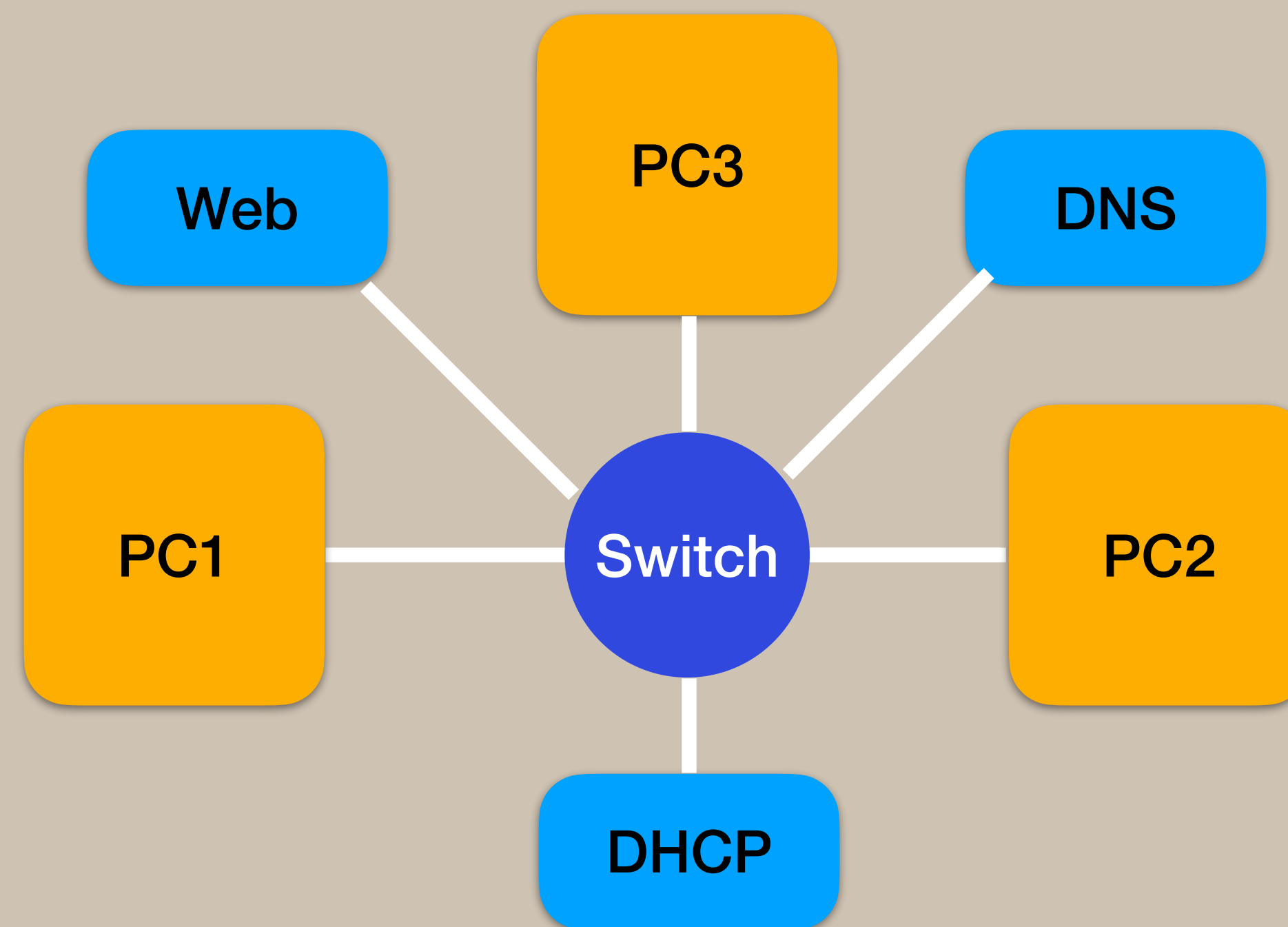
Local Area Network (LAN)



A Simple LAN of 3 PC(s)

Local Area Network (LAN)

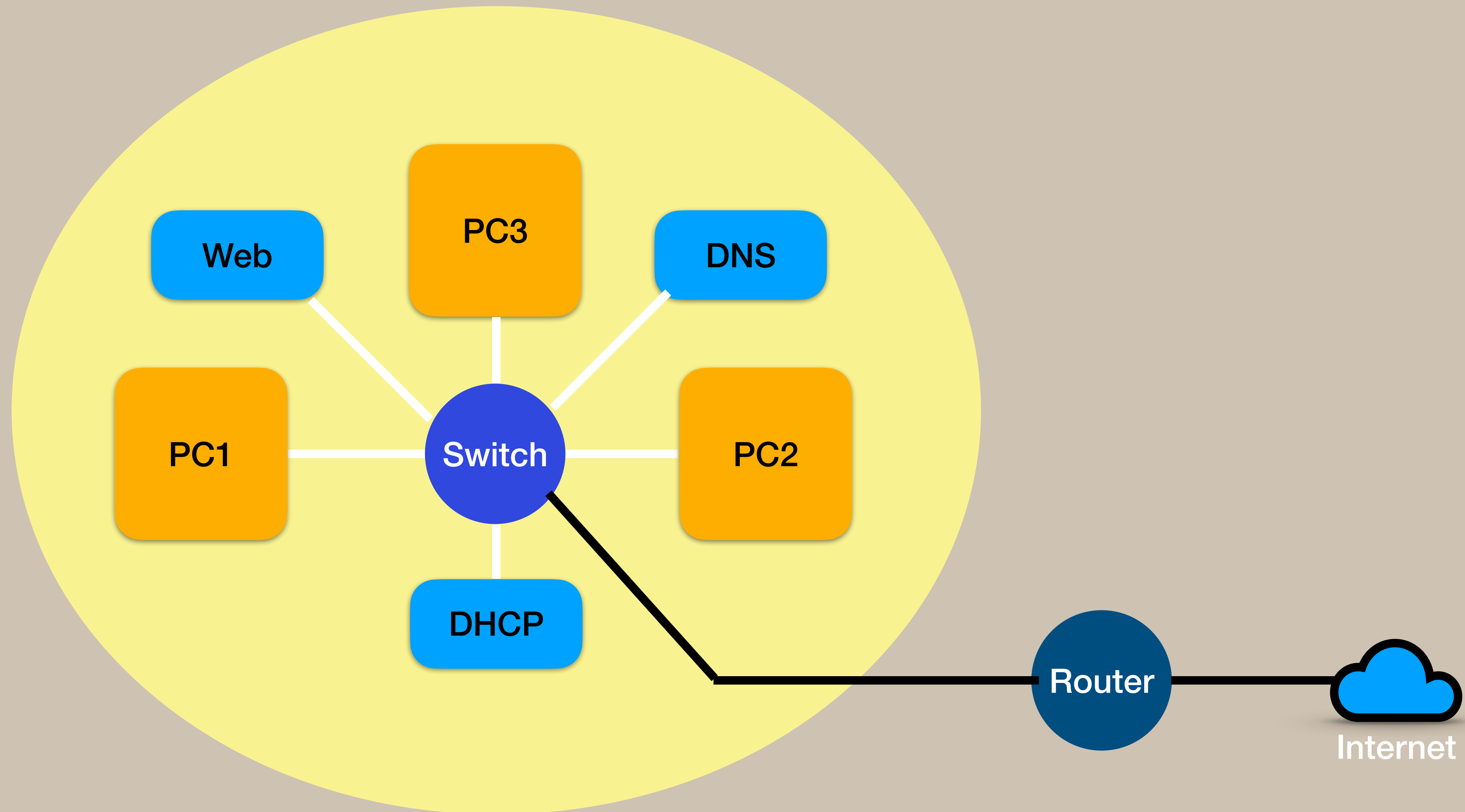
Collection of end devices **and servers** connected together via network switches and routers



A Star Topology LAN

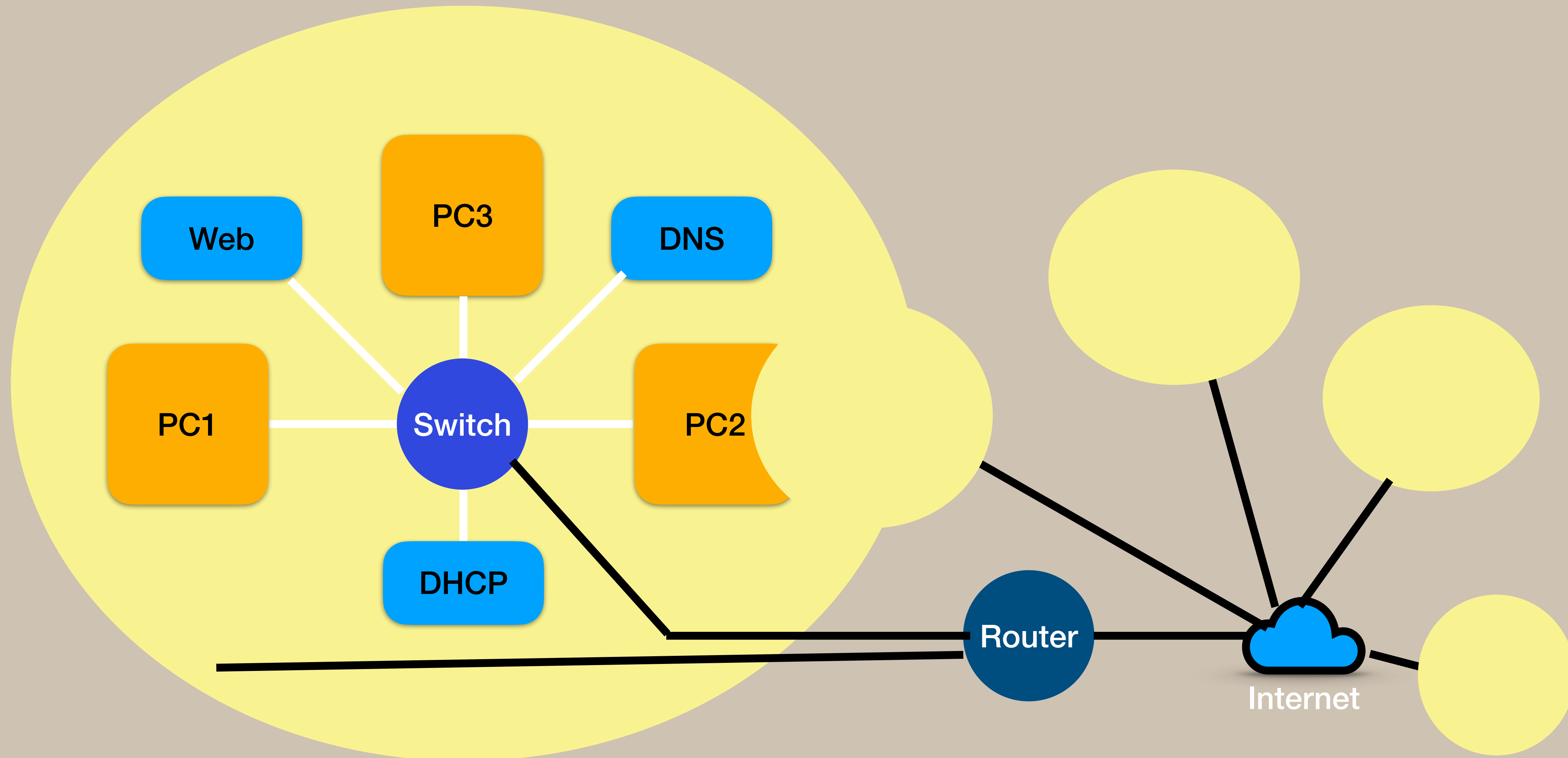
Local Area Network (LAN)

Collection of end devices **and servers** connected together via network switches and **routers**



Local Area Network (LAN)

Collection of end devices **and servers** connected together via network switches and **routers**



Networking Devices

Router

- To connect multiple networks together
- No router, no Internet
- Also give dynamic IP address to devices
- Most expensive

Switch

- To connect devices within a LAN together
- Switch only understands MAC address
- Can be 8 ports, 16 ports, 24 ports or more
- Comparably cheaper than a router with same # of ports
- Better network security compare to hub

Hub

- To connect multiple devices to form a LAN
- Does not understand either IP and MAC
- Only broadcast packets
- Use as signal amplifier (cheap)

Cables and Connections

Serial Cable

A serial cable is a cable used to connect devices that use a serial communication protocol.

This type of cable typically has a 9-pin or 25-pin connector on each end and is used to connect devices such as computers, printers, and other peripherals.

The cable transmits data one bit at a time over a single communication line or channel. Serial cables are typically used to connect devices close to each other, such as on the same desk or in the same room.

Often used to connect router of an enterprise network to the Internet

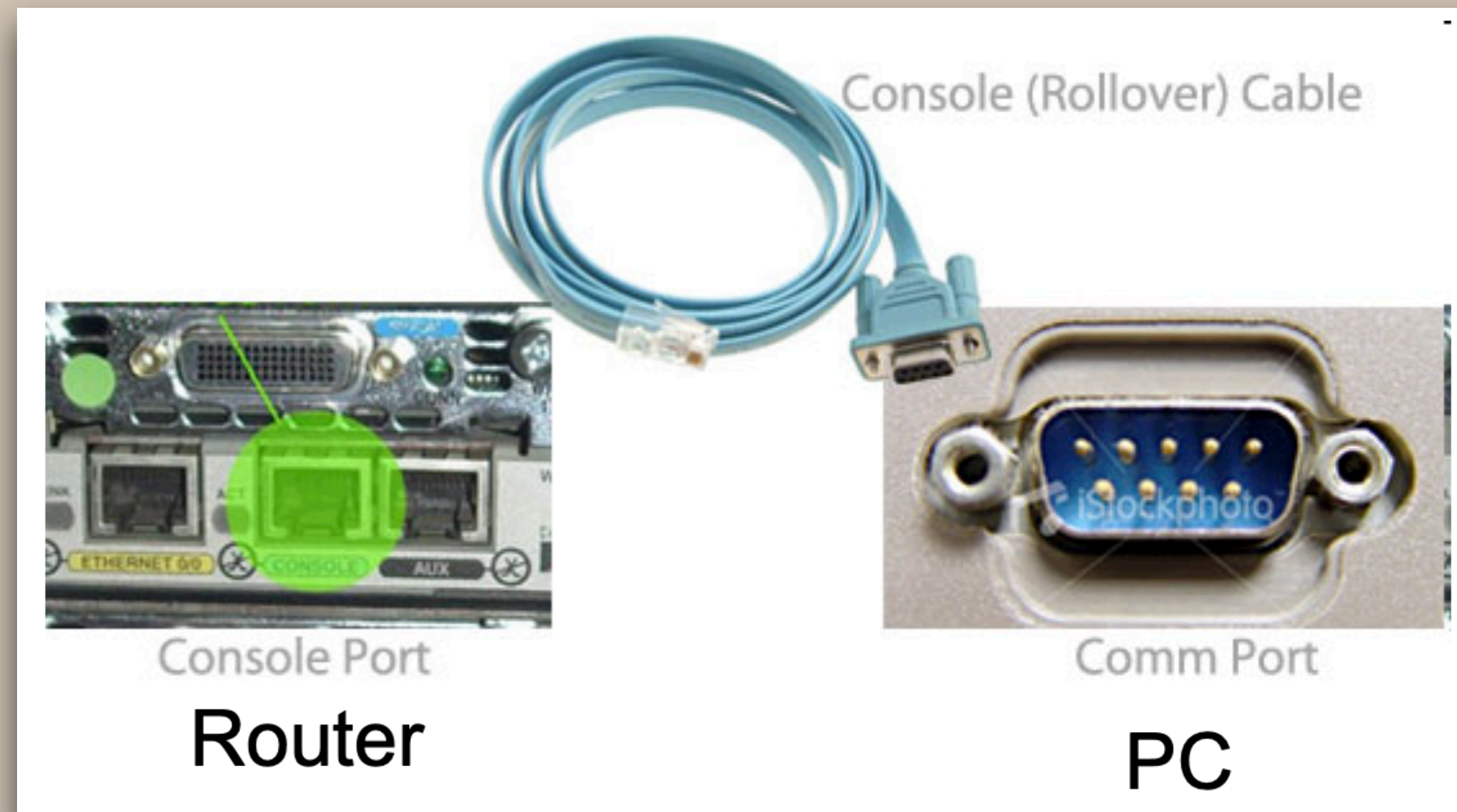


Console Cable (Roll-over Cable)

A console cable is a cable used to connect a computer or other device to the console port of a router or other network device. The console port is typically used for the initial configuration and troubleshooting of the device.

The cable has a serial comm connection at one end and a RJ-45 connector at the other end.

Rollover cable is only used for router configuration, not data transfer in network



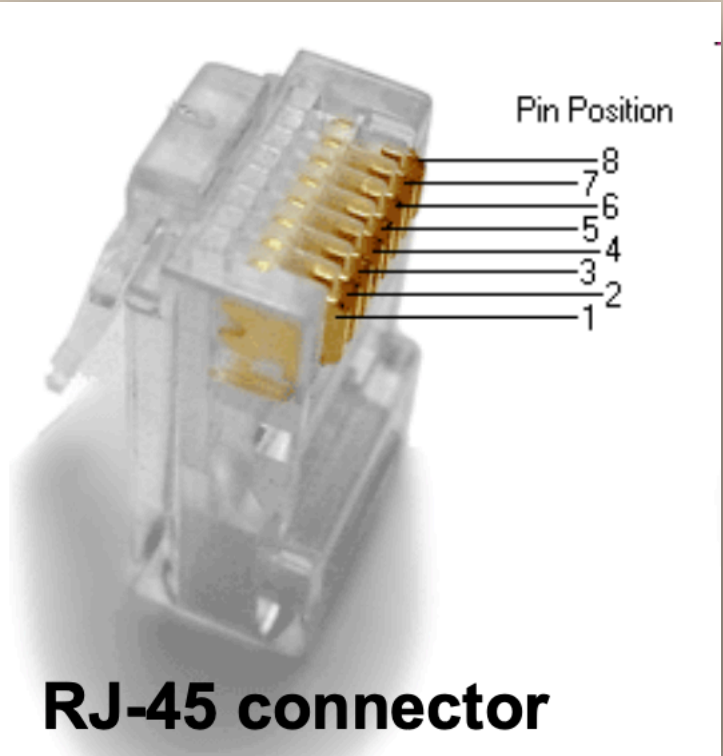
Ethernet Cables (CAT 5/6/7/8)

Ethernet cables, also known as Cat cables, are cables that are used to connect devices to a network (using RJ45 connector)

They are commonly used to connect computers, servers, switches, routers, and other network devices together.

Ethernet cables come in a variety of types and are classified by their category or Cat rating.

Category	Max speed transmission	Max bandwidth
Cat 5	100 Mbps	100 MHz
Cat 5e	1,000 Mbps	100 MHz
Cat 6	1,000 Mbps	250 MHz
Cat 6a	10,000 Mbps	500 MHz
Cat 7	10,000 Mbps	600 MHz
Cat 7a	10,000 Mbps	1,000 MHz



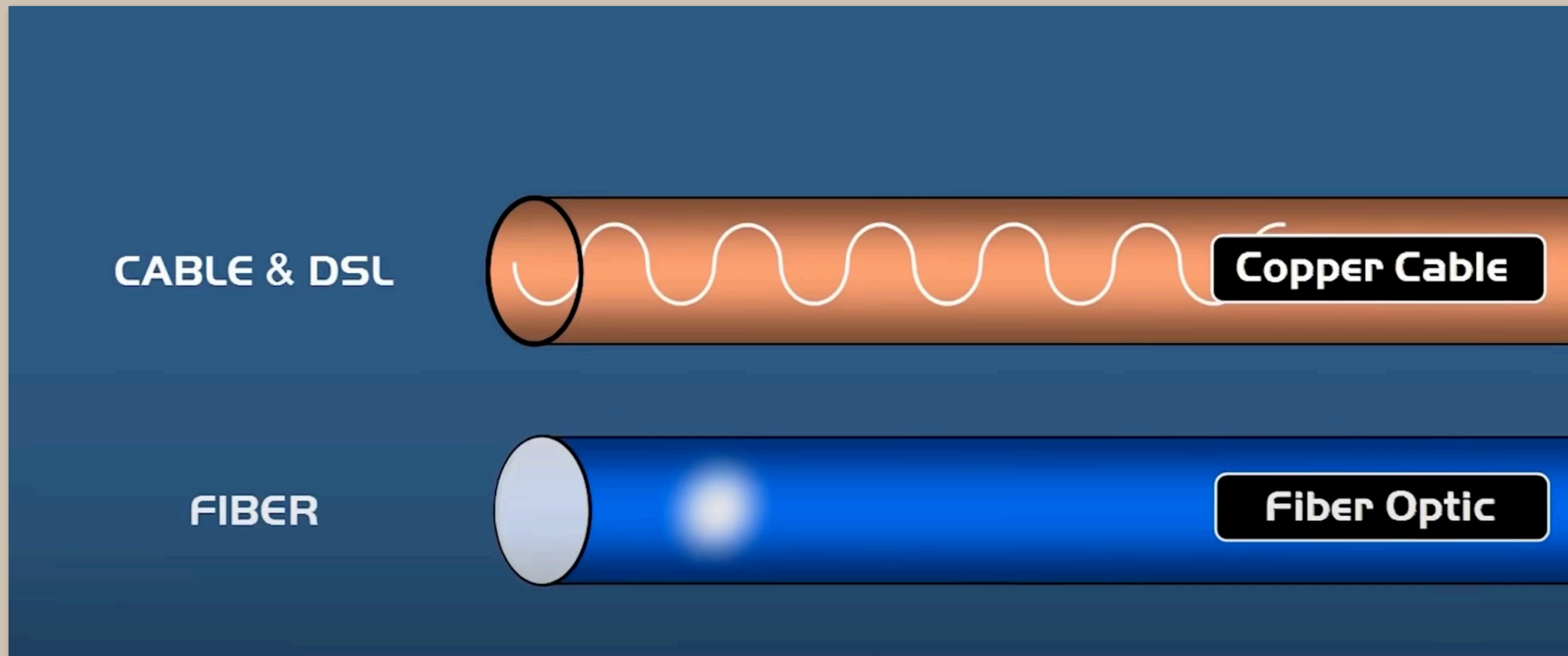
RJ-45 connector

Fiber Optics

Optical fibers are used to transmit data using light, which permits transmission over longer distances and at higher bandwidths (data rates) than other forms of communications.

Undersea Internet cables are almost all in fiber-optics

- Transmit up to tens of gigabits per second.
- Immune to electro-interference (but not shark)

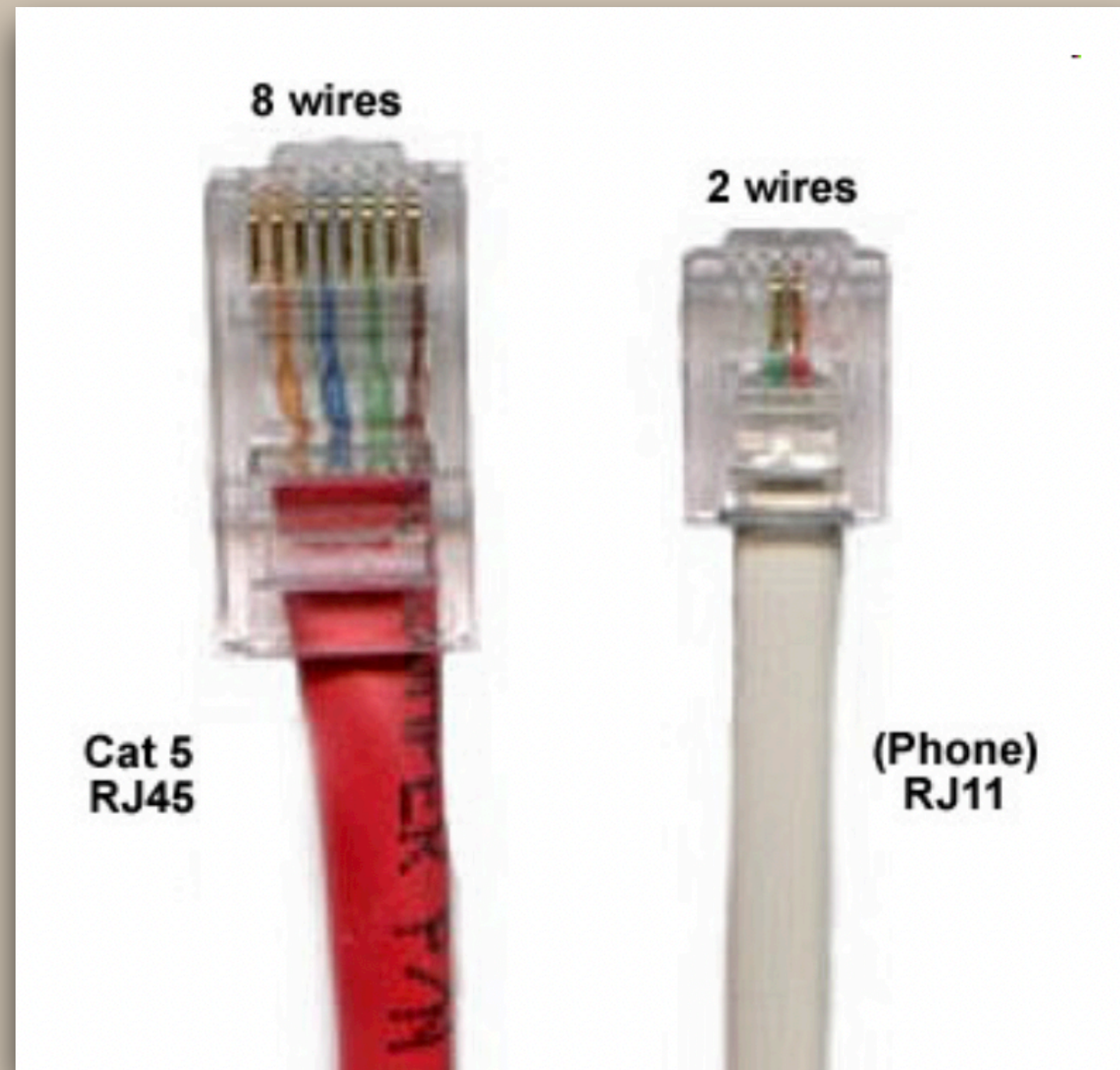


Telephone Cable

Used in ADSL modem to connect to phone plug for accessing the Internet.

For dial up connection; users need to establish a connection to server before using the Internet

Run on traditional phone lines



Router

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- No router, no Internet
- Also give dynamic IP address to devices
- Most expensive

Switch

- To connect devices within a LAN together
- Switch only understands MAC address
- Can be 8 ports, 16 ports, 24 ports or more
- Comparably cheaper than a router with same # of ports
- Better network security compare to hub

Switch connects all devices in a LAN together

A switch operates by maintaining a table (known as a forwarding table or MAC address table) that maps the MAC addresses of devices on the network to their corresponding network ports.

When a data packet arrives at the switch, it examines the destination MAC address in the packet's header and uses the forwarding table to determine which port the packet should be sent to.

The switch then forwards the packet only to that specific port, rather than broadcasting it to all connected devices as a hub would.

Hub

- To connect multiple devices to form a LAN
- Does not understand either IP and MAC
- Only broadcast packets
- Use as signal amplifier (cheap)

Hub as signal amplifier

A hub can be used as a repeater.

Hubs were commonly used in the early days of networking as a way to connect multiple devices in a network, but they have largely been replaced by switches, which provide better performance and more advanced features.

A repeater simply amplifies the signal it receives, allowing it to be transmitted over a longer distance, but it does not have the ability to selectively forward data to specific devices in the network like switches do.

Often used to source and destinations that are very far apart

DEMO (When to use hub or switch)

Hub vs Switch

Hub is a broadcast devices. In 'hub' networking, there are more chances of packet collisions which affects network performance. There is also security concerns since the packets are sent to all devices in the LAN

Switch forwards data frames based on MAC address of the destination hosts. This means that the frame is only sent from one sender to one receiver (Unicast). Thus, there is less data collisions and the data privacy is preserved.

When to use Switch or Router?

Router vs Switch

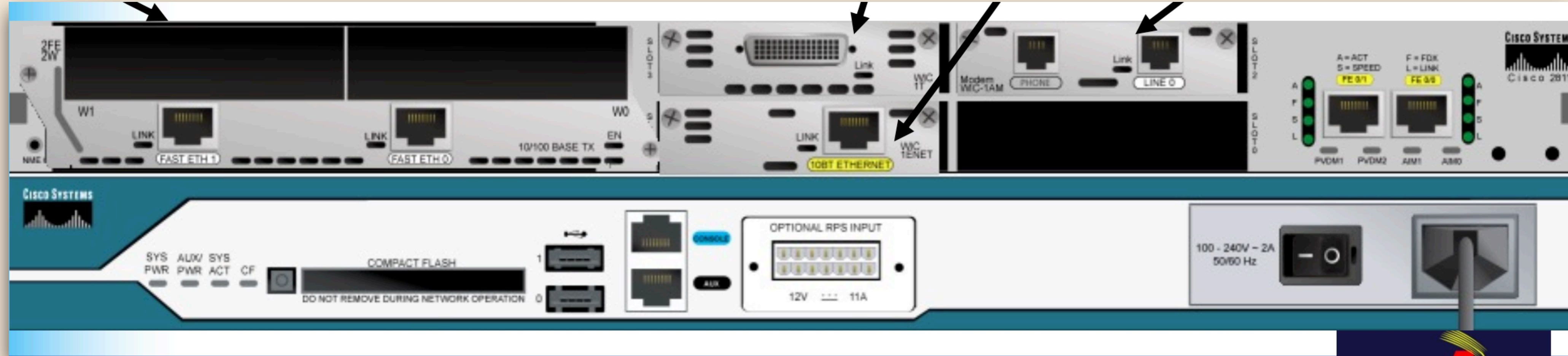
Routers are used to route data packets between multiple networks, while switches are used to provide a connection point for devices within a single network.

Routers operate at the Network Layer (Layer 3), while switches operate at the Data Link Layer (Layer 2).

Routers divide broadcast domains, while switches maintain broadcast domains.

Switches are generally faster than routers because they operate at a lower layer of the OSI model and are specifically designed to handle data transmission within a single network.

Router customization



Cisco IOS

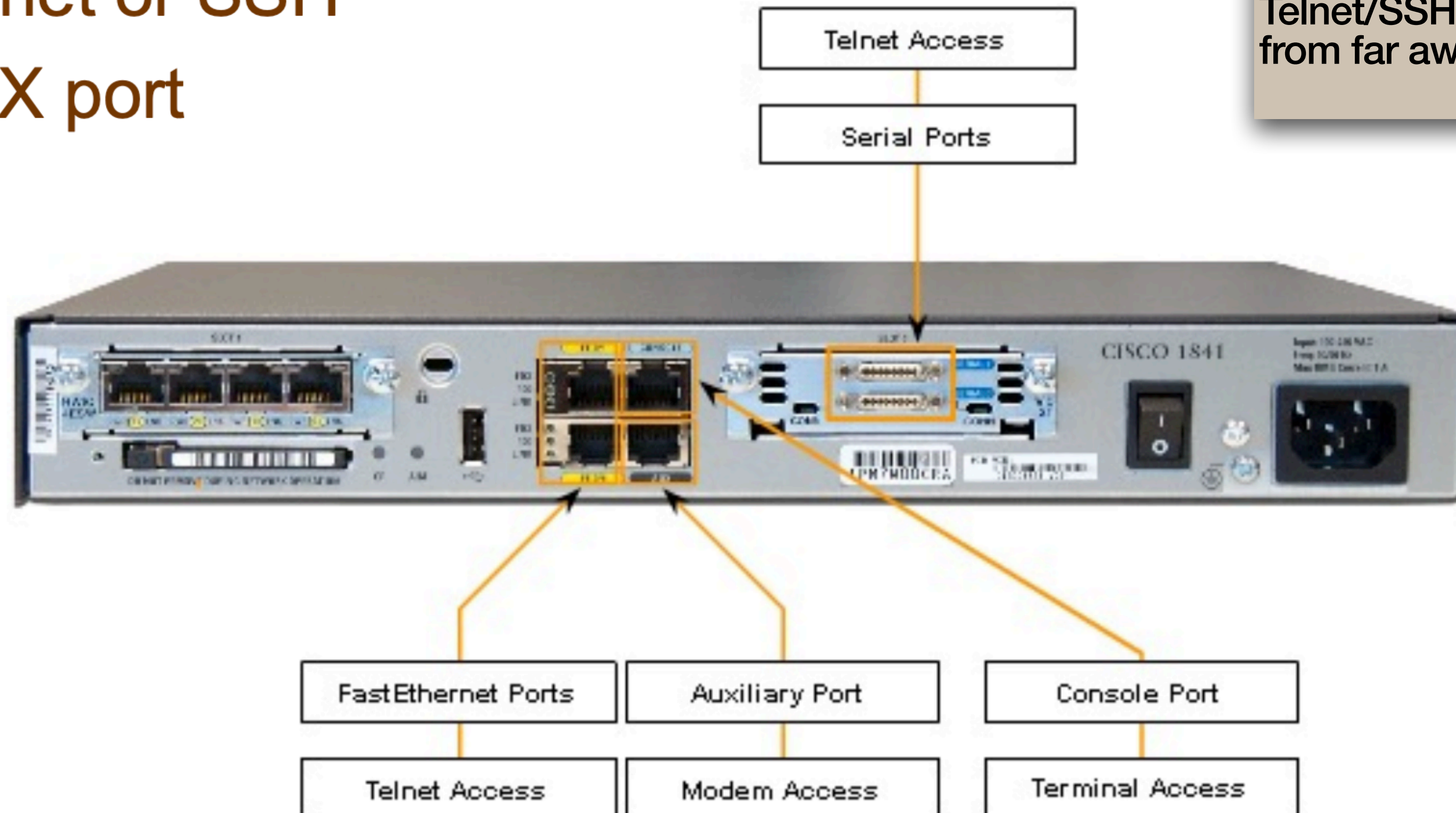
Router>	User mode
Router#	Privileged mode (also known as EXEC-level mode)
Router(config)#	Global configuration mode
Router(config-if)#	Interface mode
Router(config-subif)#	Subinterface mode
Router(config-line)#	Line mode
Router(config-router)#	Router configuration mode

How to connect to router?

- Console
- Telnet or SSH
- AUX port

Console: direct connection, use console cable

Telnet/SSH: remote login, can configure router from far away



To build a LAN, can use either HUB or SWITCH. Use Hub to cut cost, use Switch for better security

To connect one LAN to other LAN, use a router. All LANs connected to each other become the Internet

Network Topology

Network Topology

Network topology: the arrangement or mapping of the elements (links, nodes, etc.) of a network, especially the physical (real) and logical (virtual) interconnections between nodes.

Bus/Line topology: a linear network in which all devices are connected to a single cable or backbone

Star topology: a central hub or switch connects to each device on the network

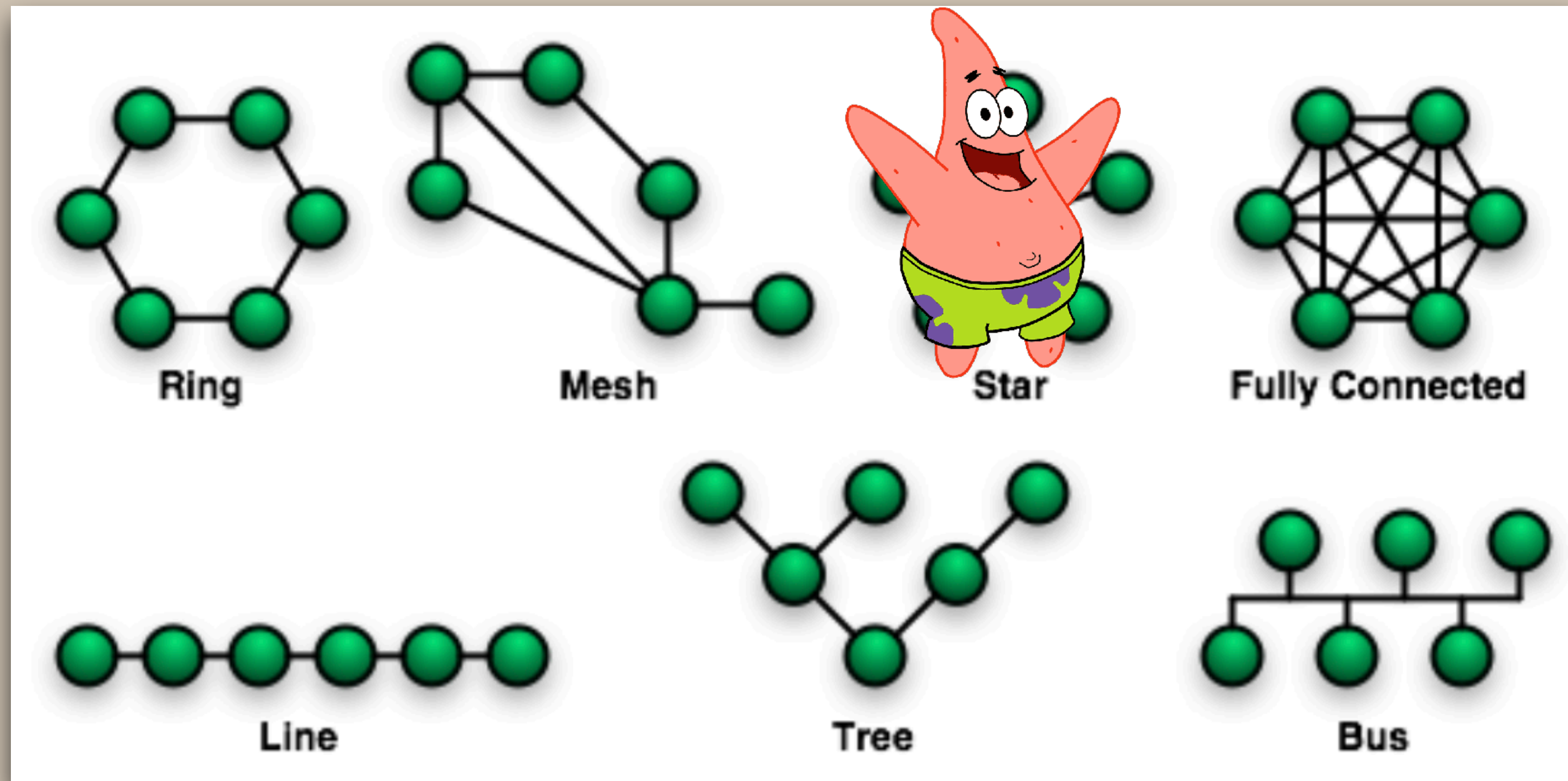
Ring topology: a closed loop in which each device is connected to two other devices, with data flowing in one direction

Mesh topology: each device is connected to every other device on the network

Tree topology: a hybrid of bus and star topologies with a central backbone connecting to multiple star networks

Hybrid topology: a combination of two or more of the above topologies

Network Topology



Network Topology DEMO